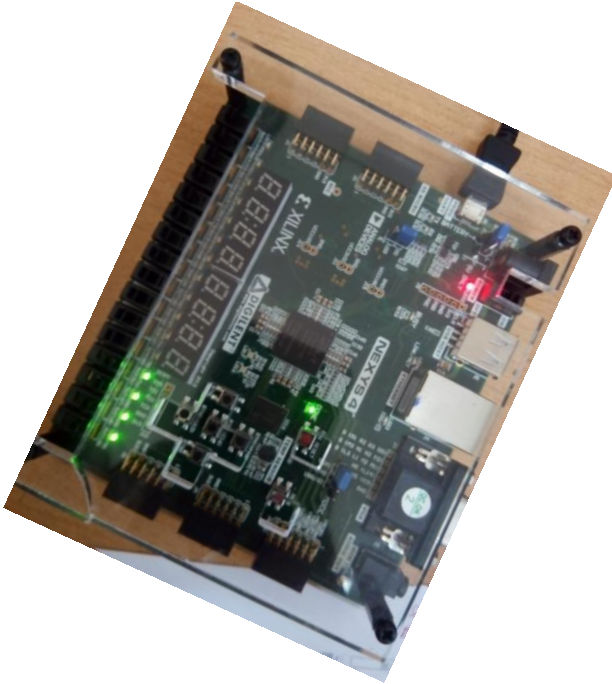




TP3

Afficheur multiplexé

REALISE PAR :

Limème Hend
ouerghemmi Sarra

classe
2 gec 1-1

Les codes vhdL necessaires :

Code 1 bits :

```
library IEEE;
use IEEE.STD_LOGIC_1164.all;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.all;

entity add1 is
    port(a,b,cin: in std_logic; s,cout: out std_logic);
end;

architecture arch1 of add1 is
begin
    s <= a xor b xor cin;
    cout <= (a and b) or (a and cin) or (b and cin) ;
end arch1;
```

code 4 bits :

```
library IEEE;
use IEEE.STD_LOGIC_1164.all;
use IEEE.STD_LOGIC_ARITH.ALL;
use IEEE.STD_LOGIC_UNSIGNED.all;

entity add4 is
    port(a,b: in std_logic_vector(3 downto 0); cin:in std_logic; s: out std_logic_vector(3 downto 0);cout: out std_logic);
end;

architecture arch1 of add4 is
    component add1
        port(a,b,cin: in std_logic; s,cout: out std_logic);
    end component;

    signal n: std_logic_vector(2 downto 0);

    begin
        p1:add1 port map (a(0),b(0),cin,s(0),n(0));
        p2:add1 port map (a(1),b(1),n(0),s(1),n(1));
        p3:add1 port map (a(2),b(2),n(1),s(2),n(2));
        p4:add1 port map (a(3),b(3),n(2),s(3),cout);
    end arch1;
```

code 7 segments:

```
library ieee;

use ieee.std_logic_1164.all;

entity afficheur is

    port (E :in std_logic_vector(3 downto 0 );s : out std_logic_vector(6 downto 0));

end afficheur;

architecture arch of afficheur is

begin

    process (E)

    begin

        case E is

            when "0000"=> s<="1000000";

            when

                "0001"=> s<="1111001";

            when

                "0010"=> s<="0100100";

            when

                "0011"=> s<="0110000";

            when

                "0100"=> s<="0011001";

            when

                "0101"=> s<="0010010";

            when

                "0110"=> s<="0000010";

            when

                "0111"=> s<="1111000";

            when

                "1000"=> s<="0000000";

            when

                "1001"=> s<="0010000";

            when

                "1010"=> s<="0001000";
```

```

        when
            "1011"=> s<="0000011";

        when
            "1100"=> s<="1000110";

        when
            "1101"=> s<="0000110";

        when
            "1110"=> s<="0001110";

        when
            "1111"=> s<="0001110";

        when others => s<="1111111";

    end case;

```

```
end process;
```

```
end arch;
```

code 7 segmants modifié :

```
library ieee;
```

```
use ieee.std_logic_1164.all;
```

```
entity afficheur is
```

```
    port (a :in std_logic_vector(3 downto 0 );seg : out std_logic_vector(6 downto 0);r : out std_logic_vector(7 downto 0));
```

```
end afficheur;
```

```
architecture arch of afficheur is
```

```
begin
```

```
    r<="11111110";
```

```
    process (a)
```

```
    begin
```

```
        case a is
```

```
            when "0000"=> seg<="1000000";
```

```
            when
```

```
                "0001"=> seg<="1111001";
```

```
            when
```

```
                "0010"=> seg<="0100100";
```

```
            when
```

```
        "0011"=> seg<="0110000";

    when

        "0100"=> seg<="0011001";

    when

        "0101"=> seg<="0010010";

    when

        "0110"=> seg<="0000010";

    when

        "0111"=> seg<="1111000";

    when

        "1000"=> seg<="0000000";

    when

        "1001"=> seg<="0010000";

    when

        "1010"=> seg<="0001000";

    when

        "1011"=> seg<="0000011";

    when

        "1100"=> seg<="1000110";

    when

        "1101"=> seg<="0000110";

    when

        "1110"=> seg<="0001110";

    when

        "1111"=> seg<="0001110";

        when others => seg<="1111111";

    end case;

end process;

end arch;
```

code de systeme :

```
library ieee;

use ieee.std_logic_1164.all;

use ieee.std_logic_unsigned.all;

use work.all;

entity tp3 is

port(clkin : in std_logic ; a,b: in std_logic_vector(3 downto 0); cin : in std_logic ; AN3,AN2,AN1,AN0: inout std_logic; aff:out
std_logic_vector(6 downto 0) );

end tp3;

architecture arch1 of tp3 is

component afficheur

port(E: in std_logic_vector(3 downto 0);s: out std_logic_vector(6 downto 0));

end component;

component add4

port(a,b:in std_logic_vector(3 downto 0); cin : in std_logic ; s:out std_logic_vector(3 downto 0);cout:out std_logic);

end component;

signal ctr: std_logic_vector(6 downto 0);

signal a_7,b_7,s_7,c_7: std_logic_vector(6 downto 0);

signal c: std_logic;

signal c_int,s: std_logic_vector(3 downto 0);

begin

v1:afficheur port map(a, a_7);

v2:afficheur port map(b, b_7);

u1: add4 port map(a,b,cin,s,c);

c_int<="000"&c;

v3:afficheur port map(s,s_7);

v4:afficheur port map(c_int,c_7);

process(clkin)

begin

if(clkin'event and clkin='1')then

if(ctr="0000000")then

if(AN0='0')then AN0<='1'; aff<=a_7;AN1 <='0';
```

```

elsif (AN1='0') then AN1<='1'; aff<=b_7;AN2 <='0';

elsif (AN2='0') then AN2<='1'; aff<=s_7;AN3 <='0';

elsif (AN3='0') then AN3<='1'; aff<=c_7;AN0 <='0';

end if;

end if;

CTR <= CTR +"0000001";

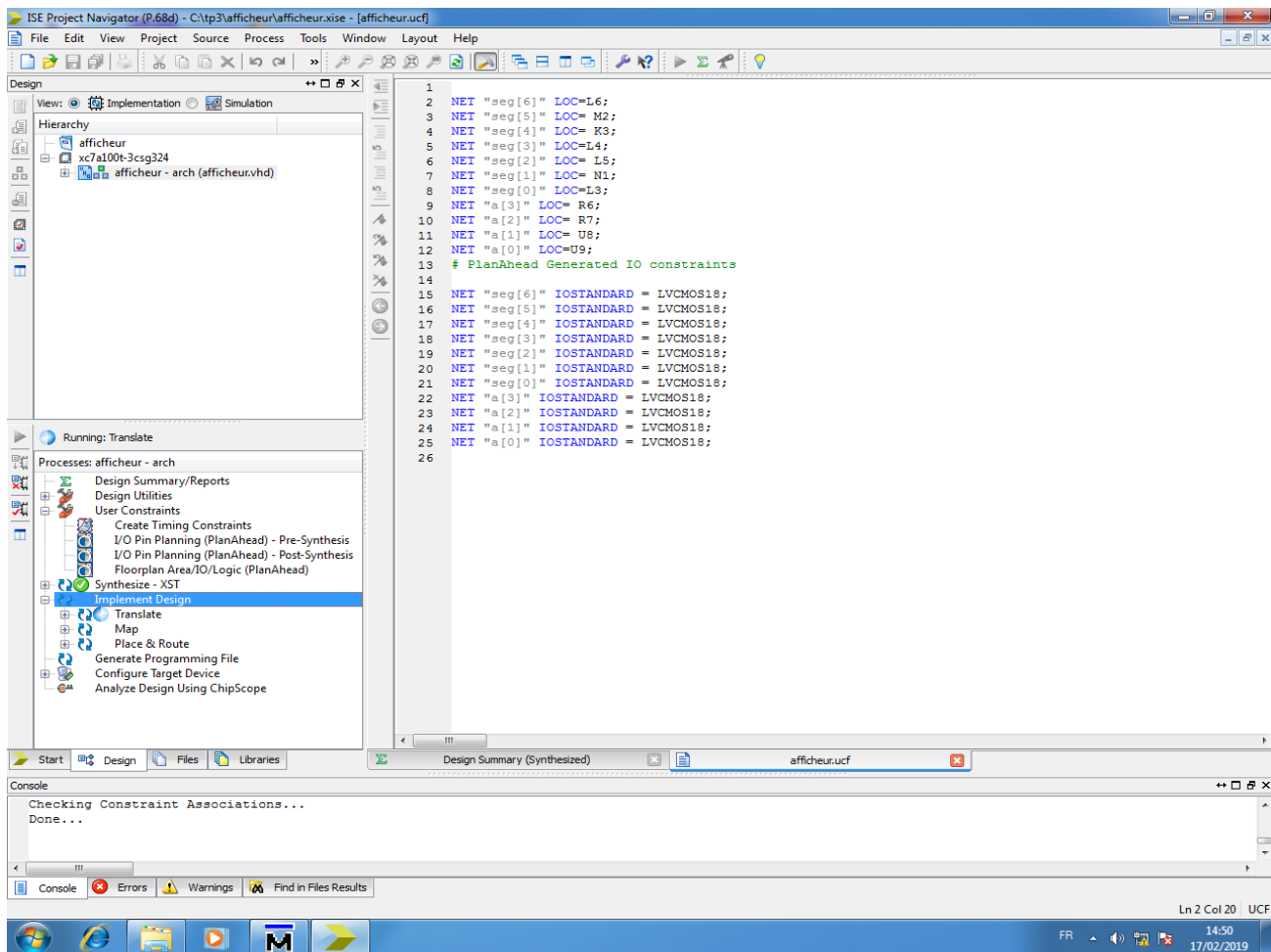
if (CTR>"1000000") then CTR <= "0000000";

end if ; end if;

end process ; end arch1;

```

implementation sur la carte fpga



ISE Project Navigator (P.68d) - C:\tp3\afficheur\afficheur.xise - [afficheur.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- afficheur
 - xc7a100t-3csg324
 - afficheur - arch (afficheur.vhd)

Running: Generate Bitstream

Processes: afficheur - arch

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead) - Pre-Synthesis
- I/O Pin Planning (PlanAhead) - Post-Synthesis
- Floorplan Area/IO/Logic (PlanAhead)
- Synthesis - XST
- Implement Design
- Translate
- Map
- Place & Route
- Generate Programming File
- Configure Target Device
- Generate Target PROM/ACE File
- Manage Configuration Project (IMPACT)
- Analyze Design Using ChipScope

```

1  NET "seg[6]" LOC=L6;
2  NET "seg[5]" LOC= M2;
3  NET "seg[4]" LOC= K3;
4  NET "seg[3]" LOC=L4;
5  NET "seg[2]" LOC= L5;
6  NET "seg[1]" LOC= N1;
7  NET "seg[0]" LOC=L3;
8  NET "a[3]" LOC= R6;
9  NET "a[2]" LOC= R7;
10 NET "a[1]" LOC= U8;
11 NET "a[0]" LOC=U9;
12 # PlanAhead Generated IO constraints
13
14
15 NET "seg[6]" IOSTANDARD = LVCMOS18;
16 NET "seg[5]" IOSTANDARD = LVCMOS18;
17 NET "seg[4]" IOSTANDARD = LVCMOS18;
18 NET "seg[3]" IOSTANDARD = LVCMOS18;
19 NET "seg[2]" IOSTANDARD = LVCMOS18;
20 NET "seg[1]" IOSTANDARD = LVCMOS18;
21 NET "seg[0]" IOSTANDARD = LVCMOS18;
22 NET "a[3]" IOSTANDARD = LVCMOS18;
23 NET "a[2]" IOSTANDARD = LVCMOS18;
24 NET "a[1]" IOSTANDARD = LVCMOS18;
25 NET "a[0]" IOSTANDARD = LVCMOS18;
26

```

Design Summary (running) afficheur.ucf

Console

generic command-line value of "Auto". Alternately, this message appears if the same option is specified multiple times on the command-line. In this case, the option listed last will be used.

Ln 2 Col 20 UCF

14:52 17/02/2019

ISE Project Navigator (P.68d) - C:\tp3\vp3\vp3.vhd - [tp3.vhd]

File Edit View Project Source Process Tools Window Layout Help

Files

Name	Type	View Association	Source
add1...	VH...	All	work
add4...	VH...	All	work
affi...	VH...	All	work
tp3.ucf	Im...	Implementation	work
tp3.v...	VH...	All	work

```

1  library ieee;
2  use ieee.std_logic_1164.all;
3  use ieee.std_logic_unsigned.all;
4  use work.all;
5  entity tp3 is
6  port(clkin : in std_logic ; a,b: in std_logic_vector(3 downto 0); cin : in std_logic ; AN3,AN2,AN1,AN0: inout std_
7  end tp3;
8  architecture arch1 of tp3 is
9  component afficheur
10 port(E: in std_logic_vector(3 downto 0);s: out std_logic_vector(6 downto 0));
11 end component;
12 component add4
13 port(a,b:in std_logic_vector(3 downto 0); cin : in std_logic ; s:out std_logic_vector(3 downto 0);cout:out std_log
14 end component;
15 signal ctr: std_logic_vector(6 downto 0);
16 signal a_7,b_7,s_7,c_7: std_logic_vector(6 downto 0);
17 signal c: std_logic;
18 signal c_int,s: std_logic_vector(3 downto 0);
19 begin
20 v1:afficheur port map(a, a_7);
21 v2:afficheur port map(b, b_7);
22 u1: add4 port map(a,b,cin,s,c);
23 c_int<="000"&c;
24 v3:afficheur port map(s,s_7);
25 v4:afficheur port map(c_int,c_7);
26 process(clkin)
27 begin
28 if(clkin'event and clkin='1')then
29 if(ctr="0000000")then
30 if(AN0='0')then AN0<='1'; aff<=a_7;AN1 <='0';
31 elsif (AN1='0') then AN1<='1'; aff<=b_7;AN2 <='0';
32 elsif (AN2='0') then AN2<='1'; aff<=a_7;AN3 <='0';
33 elsif (AN3='0') then AN3<='1'; aff<=c_7;AN0 <='0';
34 end if;
35 end if;
36 CTR <= CTR + "00000001";
37 if (CTR>"10000000") then CTR <= "00000000";
38 end if ; end if;
39 end process ; end arch1;

```

Design Summary (Programming File Generated) tp3.vhd tp3.ucf afficheur.vhd

Console

Process "Configure Target Device" launched successfully

Started : "Launching ISE Text Editor to edit afficheur.vhd".

Ln 8 Col 1 VHDL

13:56 03/03/2019

ISE Project Navigator (P.68d) - C:\tp3\vp3\tp3\vp3\tp3.xise - [tp3.ucf]

File Edit View Project Source Process Tools Window Layout Help

Files

Name	Type	View Association	Source
add1...	VH...	All	work
add4...	VH...	All	work
affic...	VH...	All	work
tp3.ucf	Im...	Implementation	work
tp3.v...	VH...	All	work

```
1 NET "a[3]" LOC=R6;
2 NET "a[2]" LOC=R7;
3 NET "a[1]" LOC=U8;
4 NET "a[0]" LOC=U9;
5 NET "aff[6]" LOC=L6;
6 NET "aff[5]" LOC=M2;
7 NET "aff[4]" LOC=K3;
8 NET "aff[3]" LOC=L4;
9 NET "aff[2]" LOC=L5;
10 NET "aff[1]" LOC=N1;
11 NET "aff[0]" LOC=L3;
12 NET "b[3]" LOC=V5;
13 NET "b[2]" LOC=V6;
14 NET "b[1]" LOC=V7;
15 NET "b[0]" LOC=R5;
16 NET "AN0" LOC=M6;
17 NET "AN1" LOC=M6;
18 NET "AN2" LOC=M3;
19 NET "AN3" LOC=N5;
20 NET "cin" LOC=V4;
21 NET "clkin" LOC=f4;
22 # PlanAhead Generated IO constraints
23
24 NET "a[3]" IOSTANDARD = LVCMOS18;
25 NET "a[2]" IOSTANDARD = LVCMOS18;
26 NET "a[1]" IOSTANDARD = LVCMOS18;
27 NET "a[0]" IOSTANDARD = LVCMOS18;
28 NET "aff[6]" IOSTANDARD = LVCMOS18;
29 NET "aff[5]" IOSTANDARD = LVCMOS18;
30 NET "aff[4]" IOSTANDARD = LVCMOS18;
31 NET "aff[3]" IOSTANDARD = LVCMOS18;
32 NET "aff[2]" IOSTANDARD = LVCMOS18;
33 NET "aff[1]" IOSTANDARD = LVCMOS18;
34 NET "aff[0]" IOSTANDARD = LVCMOS18;
35 NET "b[3]" IOSTANDARD = LVCMOS18;
36 NET "b[2]" IOSTANDARD = LVCMOS18;
37 NET "b[1]" IOSTANDARD = LVCMOS18;
38 NET "b[0]" IOSTANDARD = LVCMOS18;
39 NET "AN0" IOSTANDARD = LVCMOS18;
40 NET "AN1" IOSTANDARD = LVCMOS18;
41 NET "AN2" IOSTANDARD = LVCMOS18;
42 NET "AN3" IOSTANDARD = LVCMOS18;
43 NET "cin" IOSTANDARD = LVCMOS18;
44 NET "clkin" IOSTANDARD = LVCMOS18;
```

Start Design Files Libraries

Design Summary (Programming File Generated) tp3.vhd tp3.ucf afficheur.vhd

Console

Process "Configure Target Device" launched successfully

Started : "Launching ISE Text Editor to edit afficheur.vhd".

Ln 41 Col 33 UCF

FR 13:56 03/03/2019

ISE Project Navigator (P.68d) - C:\tp3\vp3\tp3\vp3\tp3.xise - [tp3.ucf]

File Edit View Project Source Process Tools Window Layout Help

Design

View: Implementation Simulation

Hierarchy

- vp3
- xc7a100t-3csg324
- tp3 - arch1 (tp3.vhd)

No Processes Running

Processes: tp3 - arch1

- Design Summary/Reports
- Design Utilities
- User Constraints
- Create Timing Constraints
- I/O Pin Planning (PlanAhead...)
- I/O Pin Planning (PlanAhead...)
- Floorplan Area/IO/Logic (Pla...
- Synthesize - XST
- Implement Design
- Translate
- Map
- Place & Route
- Generate Programming File
- Configure Target Device
- Analyze Design Using ChipScope

```
1 NET "a[3]" LOC=R6;
2 NET "a[2]" LOC=R7;
3 NET "a[1]" LOC=U8;
4 NET "a[0]" LOC=U9;
5 NET "aff[6]" LOC=L6;
6 NET "aff[5]" LOC=M2;
7 NET "aff[4]" LOC=K3;
8 NET "aff[3]" LOC=L4;
9 NET "aff[2]" LOC=L5;
10 NET "aff[1]" LOC=N1;
11 NET "aff[0]" LOC=L3;
12 NET "b[3]" LOC=V5;
13 NET "b[2]" LOC=V6;
14 NET "b[1]" LOC=V7;
15 NET "b[0]" LOC=R5;
16 NET "AN0" LOC=M6;
17 NET "AN1" LOC=M6;
18 NET "AN2" LOC=M3;
19 NET "AN3" LOC=N5;
20 NET "cin" LOC=V4;
21 NET "clkin" LOC=f4;
22 # PlanAhead Generated IO constraints
23
24 NET "a[3]" IOSTANDARD = LVCMOS18;
25 NET "a[2]" IOSTANDARD = LVCMOS18;
26 NET "a[1]" IOSTANDARD = LVCMOS18;
27 NET "a[0]" IOSTANDARD = LVCMOS18;
28 NET "aff[6]" IOSTANDARD = LVCMOS18;
29 NET "aff[5]" IOSTANDARD = LVCMOS18;
30 NET "aff[4]" IOSTANDARD = LVCMOS18;
31 NET "aff[3]" IOSTANDARD = LVCMOS18;
32 NET "aff[2]" IOSTANDARD = LVCMOS18;
33 NET "aff[1]" IOSTANDARD = LVCMOS18;
34 NET "aff[0]" IOSTANDARD = LVCMOS18;
35 NET "b[3]" IOSTANDARD = LVCMOS18;
36 NET "b[2]" IOSTANDARD = LVCMOS18;
37 NET "b[1]" IOSTANDARD = LVCMOS18;
38 NET "b[0]" IOSTANDARD = LVCMOS18;
39 NET "AN0" IOSTANDARD = LVCMOS18;
40 NET "AN1" IOSTANDARD = LVCMOS18;
41 NET "AN2" IOSTANDARD = LVCMOS18;
42 NET "AN3" IOSTANDARD = LVCMOS18;
43 NET "cin" IOSTANDARD = LVCMOS18;
44 NET "clkin" IOSTANDARD = LVCMOS18;
```

Start Design Files Libraries

Design Summary (Programming File Generated) tp3.vhd tp3.ucf afficheur.vhd

Console

Process "Configure Target Device" launched successfully

Started : "Launching ISE Text Editor to edit afficheur.vhd".

Ln 41 Col 33 UCF

FR 13:56 03/03/2019

